

# VELOCITY RETAIL

Autonomous AI-Driven Merchandising Engine & 3D Interactive Digital Twin Showroom

Three.js 3D WebGL

AWS ECS Fargate

AWS CloudFront (SSL)

AWS Load Balancer (ALB)

React 18 + Vite

Node.js + TypeScript

MongoDB Atlas

Groq Llama 3 (70B)

Socket.IO Live Sync

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## DEVELOPER & AUTHOR

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## PROJECT TYPE

Full-Stack Cloud & 3D WebGL Capstone Project

## LIVE PRODUCTION STATUS

AWS CloudFront CDN + ECS Fargate + Vercel

## DOCUMENT VERSION & DATE

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# 1. Project Overview & Problem Statement

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## 1.1 Physical Retail – la Enna Prachina? (The Real – World Problem)

Normal-ah physical retail fashion stores-la (like Zara, H&M, Lifestyle) visual merchandising decisions ellam **manual intuition** and static PDF planogram moolama than nadakkuthu. Store associates avangalukku thonura maari dress-ah eduthu mannequins-la maatuvaanga. Athoda sales impact epdi irukku nu avangalukku theriyathu.

- **Sales Velocity Theriyathu:** Nalla vikkura fast-moving product moolai cupboard-la thungi kittu irukkum, aana slow-moving dead stock center mannequin-la space waste pannitu irukkum.
- **Cross-Selling Miss Aaguthu:** Customer oru Shirt vaangina athuku suit aagura Pants or Shoes-ah bundle panni display panna real-time transaction synergy data irukkaathu.
- **Heavy Manual Work:** Oru display look nalla irukkuma nu paaka staff physical-ah mannequins-ah thookitu showroom full-ah alayanum.

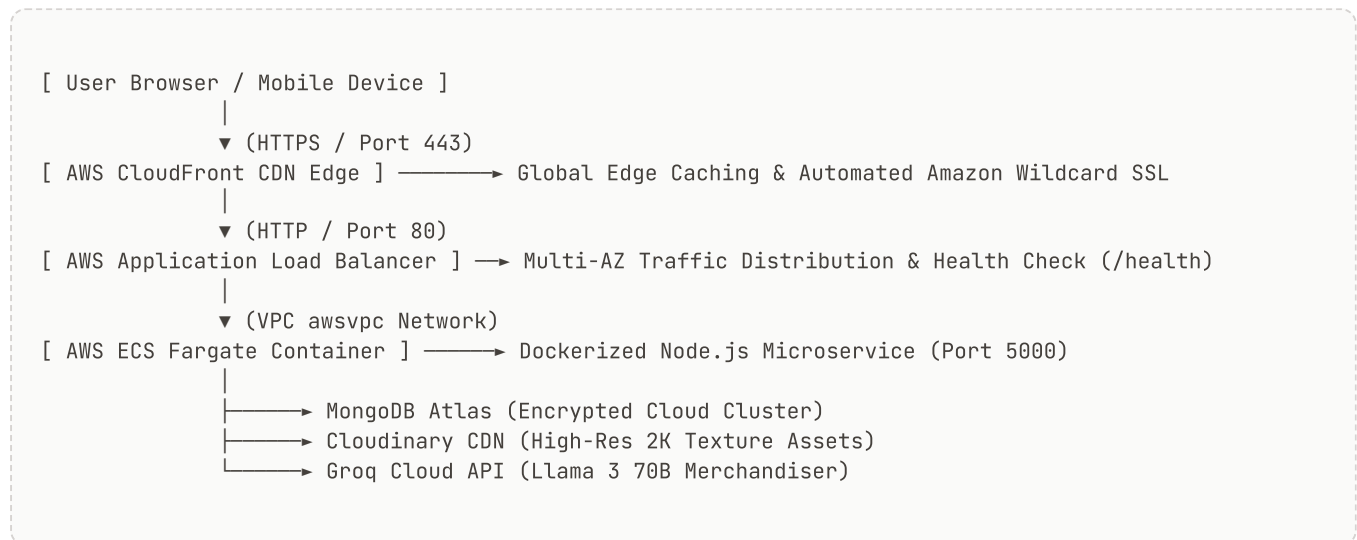
## 1.2 Namma Velocity Retail Kudukkura Solution

### Core Value Proposition

Namma project customer-oda live billing data-va analyze panni, **Apriori Algorithm (Market Basket Analysis)** and **Groq Llama 3 AI** moolama best matching outfit bundles-ah generate pannuthu. Athai oru **3D Interactive Digital Twin Showroom (Three.js WebGL)**-la store manager-kku live-ah kaati, single-click-la bilateral flight animation-la mannequins-kku deploy pannuthu!

## 2. Production Cloud Architecture (AWS-la Epdi Run Aaguthu?)

### 2.1 Complete Flow Diagram



### 2.2 Yen CloudFront Mandatory-ah Pottom? (Mentor Interview Favorite!)

ALB munnadi CloudFront podurathukku 3 mukkiyamaana technical reasons:

1. **Mixed Content Error Block Fix:** Namma frontend Vercel-la strict **HTTPS (Port 443)**-la run aaguthu. Aana ALB default-ah **HTTP (Port 80)**-la thaana irukku. Browser-la HTTPS site HTTP API-ah call panna Chrome security block pannidum. CloudFront Amazon-oda free wildcard SSL ([https://\\*.cloudfront.net](https://*.cloudfront.net)) terminate panni HTTPS enforce pannuthu.
2. **Super Fast Edge Speed:** CloudFront Chennai ( [MAA51-P3](#) ) and Bangalore ( [BLR50-P4](#) ) Edge POPs moolama requests-ah cache panni instant response kudukkum.
3. **Cookie & Header Pass-Through:** CloudFront cache behavior-la [Cookies: all](#) and [Authorization](#) , [Origin](#) headers whitelist pannirukom. Athanaala JWT session drop aagave aagathu!

# 3. 3D Digital Twin Showroom (Three.js WebGL & Bilateral Flight)

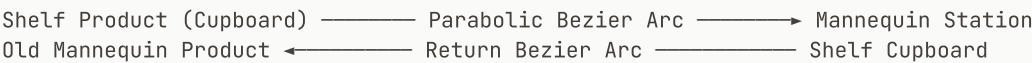
## 3.1 Showroom 3D Spatial Layout

Showroom full-ah Three.js WebGL coordinate space-la math model panni build pannirukom:

- **8 Display Cupboards:** Wall-ah suthi geometrically place aagirukku (Shirts, T-Shirts, Jeans, Jackets, Shoes).
- **2 Hero Mannequins:** Center stage-la elevated stone pedestals mela `[X: -4.5, Z: 0]` and `[X: +4.5, Z: 0]` coordinates-la spot lighting-oda fix pannirukom.

## 3.2 Bilateral 2-Way Flight Animation (Physics Math)

Normal-ah web app-la images switch aagum. Aana namma showroom-la **\*\*2-Way Simultaneous Flight\*\*** nadakkum:



Intha trajectory **Quadratic Bezier Curve** equation use panni calculate aaguthu:

$$P(t) = (1 - t)^2 \cdot P_{start} + 2(1 - t)t \cdot P_{control} + t^2 \cdot P_{end}$$

Ithula `P_control.y = max(P_start.y, P_end.y) + 3.5` nu vachu parachuted parabolic arch create panrom. Athanaala cupboards mela garments idikkaama sky-la smooth-ah fly aagi swap aagum.

## 3.3 Floating Footwear Fix (Pedestal Clearance)

### Footwear Clipping Problem & Solution

Mannequins 0.5-unit stone pedestal mela irukurathaala, shoes ground level-la render aagi pedestal kulla embedded aagi maranjuruchu. Namma vertical offset math add panni `targetSlot.y + pedestalElevation + 0.12` nu raise pannom. Ippo shoes pedestal mela Drop-Shadows-oda gilli-ya theriuthu!

## 4. AI Recommendation Engine (Apriori & Groq LLM)

### 4.1 Market Basket Analysis (MBA) Formulas

| Metric     | Math Formula                                 | Business Meaning                                                              |
|------------|----------------------------------------------|-------------------------------------------------------------------------------|
| Support    | $\text{freq}(A \cup B) / \text{Total Bills}$ | Rendu products sernthu ethana customer vangaangalo antha percentage.          |
| Confidence | $\text{freq}(A \cup B) / \text{freq}(A)$     | Item A vaangura customer Item B-ayum kandippa vaangura probability.           |
| Lift       | $\text{Confidence} / \text{Support}(B)$      | Lift > 1.0 iruntha than athu genuine complementary synergy (e.g. Suit + Tie). |

### 4.2 Groq Llama 3 (70B) AI Copilot

Apriori mathematical scores generate pannum. Aana **Groq Llama 3 AI** athai human retail strategy-ah mathi manager-kku recommend pannum:

- Slow moving dead-stock-ah best seller kooda bundle panni fast-ah liquidate panna strategy solra prompt engineering.
- Real-time AI Copilot Drawer-la staff ethu pathi kettaalum inventory velocity insights instant-ah explain pannum.

## 5. Station Reset & Incremental Stacking Bug Fix

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### 5.1 User Kanda 2 Critical Bugs

1. **1-by-1 Overwrite Bug:** Oru station-la 4 items-ah onnu onna swap pannitu, apram Reset Station click panna 1 item mattum shelf-ku poguthu, matha 3 mannequin-laye ninnuduthu.
2. **20-30 Products Sky-la Fly Aana Wildcard Bug:** Meethi 3 items-ah thirumba reset panna try panna, store full-ah ulla 20-30 cupboards boxes sky-la fly aagi floating aayiduthu.

### 5.2 Root Cause & Namma Panna Permanent Fix

#### How We Solved It Completely

1. **Backend Stacking ( `executeFloorSwap` ):** Pazhaiya code puthu item varum bothu mathatha delete pannichu. Ippo atomic merging panni station-la ulla ella roles-um (Jacket, Shirt, Pants, Shoes) preserve aagura maari stack pannitom.
2. **Wildcard Flight Guard:** Empty array ( `revertedProductIds: []` ) vantha flight animation-ah trigger panna koodathu nu `if (!specificProductIds || specificProductIds.length === 0) return;` guard potom.
3. **Scoped Flight:** Flight animation loop-la antha specific station mathi cupboards fly aagatha maari boundary restrict pannitom.

## 6. Mobile Login Issue & Dual-Layer Authentication Fix

### 6.1 Mobile-la Yen Login Aagala?

Rendu mukkiyamaana karanam:

- **Google Breach Alert Popup:** Pazhaiya password `admin123` weak-ah irundhathaala, Android Chrome OS-level-la popup pottu screen-ah block panniruchu.
- **Cross-Site Cookie Block on Mobile:** Mobile Chrome / Safari tracking protection HttpOnly session cookies-ah block panniduchu, athanaala dashboard load aana maru-nodiyame logout aagi login-kku bounce aachu.

### 6.2 Namma Panna Permanent Fix (Dual-Layer Auth)

| Auth Method           | Epdil Work Aaguthu?                                                                                                                | Result                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| HttpOnly Cookies      | Desktop browsers-la automatic-ah cookie transmit aagum                                                                             | XSS Attack Protection                               |
| Bearer Token Fallback | <code>localStorage</code> token-ah Axios Request Interceptor-la <code>Authorization: Bearer &lt;token&gt;</code> -ah attach panrom | Mobile-la cookies block aanaalum 100% smooth login! |
| Strong Credentials    | <code>Velocity@Admin2026!</code> password set pannitom                                                                             | Google Password Manager breach popup zero!          |

## 7. Database Design & MongoDB Atlas Schema Models

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Project-Ia 6 core Mongoose Schemas irukku:

- **User:** Staff credentials and RBAC roles (Admin, Manager, Staff) with bcrypt 10 rounds hash.
- **Product:** 3D showroom coordinates `{ x, y, z }`, Category, SKU, Price, Stock, Cloudinary image URL.
- **Sale:** POS billing transactions, purchased items array, total amount, cashier ID.
- **Planogram:** Active visual merchandising layouts and shelf zone mappings.
- **FloorSwap:** Station swaps, executedItems and displacedItems tracking for 2-way bilateral flight and resets.
- **CategoryVelocity:** 7-day velocity aggregation, units sold daily, turnover score.

## 8. Role-Based Access Control (RBAC) & Security

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| Role                                                                                                     | 3D Showroom | AI Recommendations    | Product Catalog      | User Management          |
|----------------------------------------------------------------------------------------------------------|-------------|-----------------------|----------------------|--------------------------|
|  <b>Admin</b>           | Full Access | Deploy & Reset        | Add, Edit, Delete    | Add, Role Change, Delete |
|  <b>Manager</b>         | Full Access | Deploy & Reset        | View Catalog & Stock | Blocked (403 Forbidden)  |
|  <b>Cashier / Staff</b> | View Only   | Read Only (No Deploy) | View Products Only   | Blocked (403 Forbidden)  |

## 9. Complete AWS Production Deployment

### Commands Runbook

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```
# 1. Docker Image Build & Push
docker build -t aaronalen/velocity-retail-backend:latest ./backend
docker push aaronalen/velocity-retail-backend:latest

# 2. Register ECS Fargate Task Definition with .env variables
aws ecs register-task-definition --cli-input-json file://task-def.json --region ap-south-1

# 3. Create Target Group with /health check
aws elbv2 create-target-group --name "velocity-retail-tg" --protocol HTTP --port 5000 --vpc-id "vp
c-039619cf26628c1dd" --target-type ip --health-check-path "/health" --region ap-south-1

# 4. Create Internet-Facing ALB
aws elbv2 create-load-balancer --name "velocity-retail-alb" --subnets subnet-011dd38b725d032a8 sub
net-0f135b19916678ab9 subnet-0943f394c63efffa2 --security-groups sg-0cb2b29a1dbfc6e87 --scheme int
ernet-facing --region ap-south-1

# 5. Create CloudFront Distribution with HTTPS Redirect & Cookies Forwarding
aws cloudfront create-distribution --distribution-config file://cf-config.json

# 6. Vercel Frontend Rewrites Route to CloudFront:
# /api/* → https://d2r14mddgufil7.cloudfront.net/api/*
```

## 10. Top 15 Mentor Viva Q&A (Tanglish Presentation Script)

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### Q1: Retail management-kku yen 3D visualizer choose panneenga?

**Answer:** Normal 2D tables-la store associates-kku visual layout feel kedaikaathu sir. Fashion retail eppovume spatial and visual experience. Three.js 3D twin use panrathala, real physical store-ah maathurathukku munnadiye floor aesthetics and customer eye-level impact-ah digital-ah simulation pannipaathukalaam.

### Q2: Neenga AWS use pannirukeenga nu epdi prove pannuveenga?

**Answer:** Chrome DevTools Network Tab open panni paatha, namma API calls direct-ah `:authority: d2r14mddgufl7.cloudfront.net` -kku pogum sir. Response headers-la `Via: 1.1 ... cloudfront.net` and `X-Amz-Cf-Pop: BLR50-P4` theriyum. Athu namma AWS ALB moolama AWS ECS Fargate container-kku route aagi data fetch pannuthu.

### Q3: WebSocket / Socket.IO yen theva pattuthu?

**Answer:** Oru store-la multiple staff tablets vachirupaanga. Manager mannequin-kku oru bundle swap panna pothu, matha ella floor terminals-layum real-time-ah bilateral flight animation fly aagi sync aaganum. Manual page refresh theva illama instant sync aaga Socket.IO gateway use pannirukom sir.